Distributed Adaptive Resource Allocation under Lipschitz Continuous Cost Functions

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***Abstract*—This paper studies the distributed adaptive resource allocation problem under cost functions with Lipschitz continuous gradients. By relaxing the traditional strong convexity assumption, we construct a cross-term Lyapunov function to establish semi-global stability of the uncertain saddle-point dynamics. Theoretical results are validated through simulations.**

***Index Terms*—Distributed optimization, adaptive control, Lipschitz continuity, Lyapunov stability, saddle-point dynamics.**

I. Introduction

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II. Preliminaries

A. Graph Theory

Consider a weighted digraph $G(\mathcal{V},\mathcal{E},W)$, where $\mathcal{V}=\{1,2,\dots,N\}$ is the node set, $\mathcal{E}\subseteq\mathcal{V}\times\mathcal{V}$ is the edge set, and $W=[w_{ij}]\in\mathbb{R}^{N\times N}$ is the weighted adjacency matrix with $w_{ij}>0$ if $(j,i)\in\mathcal{E}$ and $w_{ij}=0$ otherwise.

The associated Laplacian matrix $L=[L_{ij}]$ is defined as

$$L_{ij}=\begin{cases}-w_{ij}, & i\neq j,\\ \sum_{k=1,\,k\neq i}^{N} w_{ik}, & i=j.\end{cases} \tag{1}$$

It is well known that $L\mathbf{1}_N=\mathbf{0}_N$, and L has a simple zero eigenvalue if the digraph G contains a directed spanning tree.

Remark 1: Throughout this paper, the communication graph G is assumed to contain at least one directed spanning tree (DST). This assumption is weaker than the strong connectivity or weight-balanced conditions commonly adopted in existing works [2]. The DST condition is sufficient to ensure consensus-like convergence in adaptive dynamics, while allowing more general analysis under milder graph assumptions.

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B. Directed Spanning Tree and DST-Related Matrices

A *directed spanning tree* (DST) $\tilde{G}(\mathcal{V},\tilde{\mathcal{E}},\tilde{W})$ is a subgraph of G that contains a unique root node (without in-neighbors) and admits exactly one directed path from the root to any other node. Without loss of generality, the root is labeled as node 1. Let p_k denote the parent of node $k+1$ in $\tilde{G}$, $k=1,\dots,N-1$.

Following [2], define the incidence-like matrix $\Xi\in\mathbb{R}^{(N-1)\times N}$ as

$$\Xi_{kj}=\begin{cases}-1, & j=k+1,\\ 1, & j=p_k,\\ 0, & \text{otherwise}.\end{cases} \tag{2}$$

Then, Ξ has full row rank and satisfies

$$\Xi L=Q\Xi, \tag{3}$$

where $Q\in\mathbb{R}^{(N-1)\times(N-1)}$ depends uniquely on the DST topology.

Remark 2: The communication graph G is assumed to contain at least one DST, which is a weaker assumption than requiring G to be strongly connected or weight-balanced [2]. The matrix Q encapsulates the essential connectivity structure of the DST and will later appear in the Lyapunov function derivative, allowing stability analysis in the reduced $(N-1)$-dimensional space.

a) Spectral Properties: The eigenvalues of Q coincide with the nonzero eigenvalues of L, which implies $Q^TQ>0$ and $Q^S=(Q+Q^T)/2>0$. Consequently,

$$\underline{\lambda}(Q^S)=\lambda_{\min}^{+}(\mathcal{L}^S), \tag{4}$$

where $\lambda_{\min}^{+}(\mathcal{L}^S)$ denotes the smallest nonzero eigenvalue of $\mathcal{L}^S$.

b) DST-Based Decomposition of Q: The matrix Q can be decomposed as

$$Q=\tilde{Q}+\bar{Q}, \tag{5}$$

where $\tilde{Q}$ is the fixed DST component, and $\bar{Q}$ represents the adaptive edge-weight contribution. This decomposition will be used to separate constant and time-varying parts in the Lyapunov analysis.

C. Useful Inequalities and Lemmas

Lemma 1 (Peter–Paul Inequality [?]): For any vectors $a,b\in\mathbb{R}^n$ and any constant $\epsilon>0$, it holds that

$$a^\top b\le\frac{1}{2\epsilon}a^\top a+\frac{\epsilon}{2}b^\top b. \tag{6}$$

This inequality is a direct consequence of the Cauchy–Schwarz inequality and the arithmetic–geometric mean inequality, and is also known as the Young inequality for $p = q = 2$.

Lemma 2 (Kronecker Product Decomposition [?]): Let $U \in \mathbb{R}^{N\times N}$ be a symmetric matrix, and let $x = \mathrm{col}(x_1, \ldots, x_N)$ with $x_i \in \mathbb{R}^n$. Then, it holds that

$$x^\top (U \otimes I_n)x = \sum_{k=1}^{n} y_k^\top U y_k, \tag{7}$$

where $y_k = ([x_1]_k, [x_2]_k, \ldots, [x_N]_k)^\top$ collects the kth components of $\{x_i\}_{i=1}^N$.

III. Problem Formulation

This paper builds upon the distributed adaptive resource allocation (DARA) framework [2], which assumes that each local cost function has a strongly convex gradient. The original work suggested as future research to relax this requirement. Following this motivation, we extend the framework to the case where each local cost function has a Lipschitz continuous gradient.

Consider a network of N agents $\mathcal{V} = \{1, \ldots, N\}$ interacting over a (possibly directed) graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$. Each agent i holds a local variable $x_i \in \mathbb{R}^n$ and a continuously differentiable cost $f_i(x_i)$. The network cooperatively minimizes the total cost subject to the global resource constraint:

$$\begin{aligned} \min_{x=\mathrm{col}\{x_1,\ldots,x_N\}} \quad & f(x) \triangleq \sum_{i=1}^{N} f_i(x_i) \\ \text{s.t.} \quad & \sum_{i=1}^{N} x_i = d, \end{aligned} \tag{8}$$

where $d = \sum_i d_i$.

Assumption 1: Each local cost function f_i is continuously differentiable and strictly convex.

Under this assumption, problem (8) admits a unique optimal solution x^* and Lagrange multiplier y^* satisfying the Karush–Kuhn–Tucker (KKT) conditions:

$$\nabla f(x^*) + \mathbf{1}_N \otimes y^* = 0, \tag{9}$$
$$(\mathbf{1}_N^\top \otimes I_n)(x^* - D) = 0, \tag{10}$$

where $\nabla f(x) = \mathrm{col}(\nabla f_1(x_1), \ldots, \nabla f_N(x_N))$ and $D = \mathrm{col}(d_1, \ldots, d_N)$. These equations define the equilibrium point (x^*, y^*) of the DST-based adaptive dynamics considered later.

Extension 1: In this work, we relax Assumption 1 by allowing each f_i to possess only a Lipschitz continuous gradient rather than a strongly convex one. This relaxation broadens the applicability of the distributed adaptive resource allocation framework, while a Lyapunov-based analysis will be developed to ensure semi-global convergence under this weaker condition.

Building upon the above graph-theoretic preliminaries, we now proceed to analyze the distributed saddle-point dynamics under the mild DST assumption.

A. Primary Analysis

To solve the distributed resource allocation problem (8), consider the Lagrangian function

$$L(x, y) = f(x) + y^\top (\mathbf{1}_N^\top \otimes I_n)(x - D), \tag{11}$$

where $x = \mathrm{col}(x_1, \ldots, x_N)$ and $y \in \mathbb{R}^n$ is the dual variable. The classical centralized saddle-point dynamics are given by

$$\dot{x} = -\nabla f(x) - \mathbf{1}_N \otimes y, \tag{12}$$
$$\dot{y} = (\mathbf{1}_N^\top \otimes I_n)(x - D), \tag{13}$$

which satisfy the KKT conditions (9)–(10) at equilibrium.

To implement a fully distributed scheme, each agent i maintains a local dual variable $y_i \in \mathbb{R}^n$ and an auxiliary state $z_i \in \mathbb{R}^n$, yielding the DST-based adaptive dynamics:

$$\dot{x} = -\kappa_1 \nabla f(x) + y, \tag{14}$$
$$\dot{y} = \kappa_3(x - D) - (\mathcal{Y} \otimes I_n)y - (\mathcal{L} \otimes I_n)z, \tag{15}$$
$$\dot{z} = (\mathcal{Y} \otimes I_n)y, \tag{16}$$

where $y = \mathrm{col}(y_1, \ldots, y_N)$, $z = \mathrm{col}(z_1, \ldots, z_N)$, $\kappa_1, \kappa_3 > 0$ are design gains, and $\mathcal{L}$ is the Laplacian of the communication graph $\mathcal{G}$. The matrix $\mathcal{Y}$ captures the adaptive coupling between local dual variables, which is *a priori* unknown and will be designed to guarantee stability.

Remark 3: Here, $\kappa_1, \kappa_3 > 0$ are design gains. Compared with the original DST-based dynamics in [2], we introduce the additional gain κ_3 in (15) to allow flexible tuning in the Lyapunov-based convergence analysis. Specifically, κ_3 will be used to adjust the relative weight of the integral feedback in the dual update, facilitating semi-global stability under the relaxed Lipschitz gradient assumption.

Definition 1 (GEP point): A triple $(\tilde{x}, \tilde{y}, \tilde{z}) \in \mathbb{R}^{Nn} \times \mathbb{R}^{Nn} \times \mathbb{R}^{Nn}$ is a GEP of system (14)–(16) if, for any admissible $\mathcal{Y}$, it satisfies

$$\dot{x} = \dot{y} = \dot{z} = 0. \tag{17}$$

*Lemma 3 (*Generalized Equilibria of (4)*):* Under Assumptions 1 and 2, the uncertain system (4) admits infinitely many generalized equilibrium points (GEPs). Moreover, for any GEP $(\bar{x}, \bar{y}, \bar{z})$, the pair $(\bar{x}, \bar{y})$ satisfies

$$(\bar{x}, \bar{y}) = (x^*, \mathbf{1}_N \otimes y^*), \tag{18}$$

i.e., $\bar{x}$ solves the original resource allocation problem (8), and $(\bar{x}, \bar{y})$ is unique.

Proof 1: For any $\Upsilon \in \mathcal{M}_r^N$, the relation $(\Upsilon \otimes I_n)\bar{y} = 0$ holds, which implies $\bar{y} = \mathbf{1}_N \otimes y_0$ for some $y_0 \in \mathbb{R}^n$. Substituting $(\bar{x}, \bar{y}, \bar{z})$ into (4b) and left-multiplying by $(\mathbf{1}_N^\top \otimes I_n)$ gives

$$(\mathbf{1}_N^\top \otimes I_n)(\bar{x} - D) - (\mathbf{1}_N^\top \mathcal{L} \otimes I_n)\bar{z} = 0. \tag{19}$$

Since $\mathbf{1}_N^\top \mathcal{L} = 0$ under Assumption 2, we obtain $(\mathbf{1}_N^\top \otimes I_n)(\bar{x} - D) = 0$. Combining this with $\nabla f(\bar{x}) + \mathbf{1}_N \otimes y_0 = 0$ recovers the KKT condition (9)–(10). By Lemma 5, $(\bar{x}, \bar{y}) = (x^*, \mathbf{1}_N \otimes y^*)$ is unique.

Finally, as $\mathrm{rank}(\mathcal{L}) = N - 1$, the variable $\bar{z}$ can be shifted by any vector of the form $\mathbf{1}_N \otimes \Delta z$, producing another GEP. Therefore, there exist infinitely many GEPs distinguished by $\bar{z}$. ■

Remark 4: Lemma 6 indicates that steering the uncertain saddle-point dynamics (4) to its GEPs ensures distributed optimal resource allocation. In the subsequent sections, we will design two types of continuous adaptation laws for Υ in (4b)—one based on the DST structure and the other on node-level updates—such that the GEPs are globally attractive and the system achieves stable convergence.

IV. Main Results and Proof

A. DST-Based Adaptive Dynamics under Lipschitz Condition

We adopt the distributed adaptive dynamics proposed in [2] as the baseline structure. Specifically, consider the following system:

$$\begin{aligned}\dot{x} &= -\kappa_1\big(\nabla f(x) + y\big),\\ \dot{y} &= \kappa_3(x - D) - (\mathcal{L}^a \otimes I_n)y - (\mathcal{L}\otimes I_n)z,\\ \dot{z} &= (\mathcal{L}^a\otimes I_n)y,\end{aligned} \tag{20}$$

$$\dot{a}_{ij} = \begin{cases}\kappa_2\Big((y_j - y_i) - \displaystyle\sum_{c\in\bar{\mathcal{N}}_{\text{out}}(i)}(y_i - y_c)\Big)^{\top}(y_j - y_i), & \text{if } e_{ji}\in\overline{\mathcal{E}},\\ 0, & \text{if } e_{ji}\in\mathcal{E}\setminus\overline{\mathcal{E}}.\end{cases} \tag{21}$$

If $e_{ji}\in\overline{\mathcal{E}}$, then

$$\dot{a}_{ij} \triangleq \dot{\bar{a}}_{k+1,p_k}. \tag{22}$$

where $\kappa_1, \kappa_2 > 0$ are design gains, and $\mathcal{L}^a$ is the gain-dependent Laplacian matrix:

$$\begin{aligned}\mathcal{L}^a_{ij} &= -a_{ij}w_{ij}, \quad i\neq j,\\ \mathcal{L}^a_{ii} &= \sum_{j=1, j\neq i}^{N} a_{ij}w_{ij}, \quad i = 1,\dots,N.\end{aligned} \tag{23}$$

The adaptive gains $a_{ij}(t)$ determine the feedback strength between agents i and j, and are updated only when $e_{ji}\in\overline{\mathcal{E}}$, ensuring a distributed implementation.

Assumption 2 (Lipschitz Gradient): Each local cost function f_i has an L_i-Lipschitz continuous gradient, i.e.,

$$\|\nabla f_i(x_i) - \nabla f_i(y_i)\| \le L_i\|x_i - y_i\|, \quad \forall x_i, y_i\in\mathbb{R}^n. \tag{24}$$

Let $L = \max_i L_i$ denote the global Lipschitz constant.

Theorem 1 (Semi-Global Stability under Lipschitz Gradient): Suppose Assumption 1 and Assumption 2 hold, and the communication graph $\mathcal{G}$ contains a directed spanning tree. Then, under Assumption 3 (Lipschitz Gradient), the adaptive system (20)–(23) admits a unique equilibrium point (x^*, y^*, z^*), where $(x^*, y^*) = (x^*, \mathbf{1}_N\otimes y^*)$ satisfies the KKT conditions (9)–(10). Moreover, there exist positive constants $\epsilon_5, \epsilon_7, \alpha$ such that the equilibrium is semi-globally stable in the sense that all trajectories remain bounded and

$$\lim_{t\to\infty}(x(t), y(t)) = (x^*, \mathbf{1}_N\otimes y^*). \tag{25}$$

Remark 5: In contrast to Theorem 1 of [2], where strong convexity ensures a quadratic lower bound as in

$$\begin{aligned}(x-y)^{\top}(\bar{\mathcal{L}}^U\otimes I_n)(\nabla f(x) - \nabla f(y)) \ge\\ m(x-y)^{\top}(\bar{\mathcal{L}}^U\otimes I_n)(x-y),\end{aligned} \tag{26}$$

we only assume that ∇f is Lipschitz continuous. Consequently, the Lyapunov analysis relies on the spectral properties of $\bar{\mathcal{L}}^{\mathbf{U}}$ and on bounding $\|\nabla f(x) - \nabla f(x^*)\|$ rather than exploiting a uniform convexity constant m. This relaxation broadens the applicability of the DST-based dynamics to cost functions that are merely smooth but not necessarily strongly convex.

B. Coordinate Transformation and Disagreement Dynamics

To facilitate the stability analysis, we next introduce a coordinate transformation that eliminates the consensus subspace and reveals the essential disagreement dynamics. Define the deviation variables

$$\mu = x - \bar{x}, \quad \nu = y - \bar{y}, \quad \eta = z - \bar{z}, \tag{27}$$

$$\bar{\mu} = (\Xi\otimes\mathbf{I}_n)\mu, \quad \bar{\nu} = \kappa_3(\Xi\otimes\mathbf{I}_n)\nu, \quad \bar{\eta} = (\Xi\otimes\mathbf{I}_n)\eta. \tag{28}$$

Here, Ξ is defined as in (2). In component form, $\bar{\mu} = \text{col}(\bar{\mu}_1,\dots,\bar{\mu}_{N-1})$ with $\bar{\mu}_k = \mu_{p_k} - \mu_{k+1}$, $k\in\mathcal{I}_{N-1}$. Since $\mathcal{L}^a\in\mathcal{M}_r^N$ and $\mathcal{O}(\bar{x},\bar{y},\bar{z}) = 0$ by Definition 1, in the new coordinates (28), the dynamics of $\mathcal{O}^a$ becomes

$$\dot{\bar{\mu}} = -\kappa_1(\Xi\otimes\mathbf{I}_n)h - \kappa_1\bar{\nu}, \tag{29}$$

$$\dot{\bar{\nu}} = \kappa_3\bar{\mu} - (Q^a\otimes\mathbf{I}_n)\bar{\nu} - (Q\otimes\mathbf{I}_n)\bar{\eta}, \tag{30}$$

$$\dot{\bar{\eta}} = (Q^a\otimes\mathbf{I}_n)\bar{\nu}, \tag{31}$$

$$\dot{\bar{a}}_{k+1,p_k} = \kappa_2\left(\bar{\nu}_k - \sum_{j\in\bar{\mathcal{N}}_{\text{out}}(k+1)}\bar{\nu}_{j-1}\right)^{T}\bar{\nu}_k, \quad k\in\mathcal{I}_{N-1}. \tag{32}$$

Here $h = \nabla f(\mu + \bar{x}) - \nabla f(\bar{x})$, and Q (resp. Q^a) is defined as in Lemma 3 based on the DST $\bar{\mathcal{G}}$ and the (resp. gain-dependent) Laplacian matrix. Specifically, $Q^a = \check{Q}^a + \bar{Q}^a$, where $\check{Q}^a$ is the fixed part (since $\dot{a}_{ij} = 0$ if $e_{ji}\in\mathcal{E}\setminus\overline{\mathcal{E}}$), and $\bar{Q}^a$ is the time-varying part given by

$$\check{Q}^a_{kj} = \begin{cases}\bar{a}_{j+1,p_j}w_{j+1,p_j}, & \text{if } j = k,\\ -\bar{a}_{j+1,p_j}w_{j+1,p_j}, & \text{if } j = p_k - 1,\\ 0, & \text{otherwise.}\end{cases} \tag{33}$$

Definition (2) and properties of the Kronecker product are used in (30)–(31), and the fact that $(\Xi\otimes\mathbf{I}_n)\bar{y} = 0$ yields (32).

C. Lyapunov Function Construction and Proof

To analyze the stability of the transformed dynamics, consider the Lyapunov function

$$V = c_1V_{\bar{\mu}} + V^a_{\bar{\nu}} + c_2V_{\bar{\eta}} - \alpha\,\bar{\mu}^{\top}\bar{\nu}, \tag{34}$$

where the components are defined as

$$V_{\bar{\mu}} = \frac{1}{2}\,\bar{\mu}^{\top}\bar{\mu}, \tag{35}$$

$$V^a_{\bar{\nu}} = \frac{1}{2}\,\bar{\nu}^{\top}\bar{\nu} + \sum_{k=1}^{N-1}\frac{w_{k+1,p_k}}{2\kappa_2}\big(\bar{a}_{k+1,p_k}(t) - \phi_{k+1,p_k}\big)^2, \tag{36}$$

$$V_{\bar{\eta}} = \frac{1}{2}\,(\bar{\nu}+\bar{\eta})^{\top}(\bar{\nu}+\bar{\eta}). \tag{37}$$

The design constants c_1 and c_2 in (34) are chosen as

$$c_1 = \frac{1 + 3\,\bar{\lambda}(Q^\top Q)}{\epsilon_1\,\underline{\lambda}^2(Q^S)}, \quad c_2 = \frac{3\,\bar{\lambda}(Q^\top Q)}{\underline{\lambda}(Q^S)}, \tag{38}$$

where $\bar{\lambda}(\cdot)$ and $\underline{\lambda}(\cdot)$ denote the maximum and minimum eigenvalues, respectively.

The constant C which will be used in the following proof, is defined as

$$\kappa(\Xi^\top \Xi) = \frac{\bar{\lambda}(\Xi^\top \Xi)}{\underline{\lambda}(\Xi^\top \Xi)_{\text{nonzero}}}, \quad C = \kappa(\Xi^\top \Xi)\,L^2, \tag{39}$$

where L is the Lipschitz constant of $\nabla f(x)$.

1) Corrected derivation of $c_1\dot{V}_{\bar{\mu}}$: We start from

$$\dot{V}_{\bar{\mu}} \;=\; \bar{\mu}^\top \dot{\bar{\mu}} \;=\; -\kappa_1\,\bar{\mu}^\top\big((\Xi \otimes I_n)h + \bar{\nu}\big). \tag{40}$$

For clarity denote $B \triangleq (\Xi \otimes I_n)$. All ϵ_i below are positive constants chosen for Young-type inequalities. We bound the two inner products in (40) separately using the vector version of the Young/Cauchy–Schwarz inequality in the form

$$u^\top v \le \epsilon\|u\|^2 + \frac{1}{4\epsilon}\|v\|^2, \qquad \epsilon > 0,$$

(which follows from $(\sqrt{\epsilon}\,u - v/(2\sqrt{\epsilon}))^2 \succeq 0$).

1) Bound the gradient-related term:

$$-\kappa_1\,\bar{\mu}^\top Bh \le \kappa_1\left(\epsilon_1\|\bar{\mu}\|^2 + \frac{1}{4\epsilon_1}\|Bh\|^2\right). \tag{41}$$

2) Bound the coupling term with $\bar{\nu}$:

$$-\kappa_1\,\bar{\mu}^\top \bar{\nu} \le \kappa_1\left(\epsilon_2\|\bar{\mu}\|^2 + \frac{1}{4\epsilon_2}\|\bar{\nu}\|^2\right). \tag{42}$$

Combining (41)–(42) and multiplying by c_1 yields

$$c_1\dot{V}_{\bar{\mu}} \le c_1\kappa_1(\epsilon_1+\epsilon_2)\,\|\bar{\mu}\|^2 + \frac{c_1\kappa_1}{4\epsilon_1}\|Bh\|^2 + \frac{c_1\kappa_1}{4\epsilon_2}\|\bar{\nu}\|^2. \tag{43}$$

Next we eliminate the $\|Bh\|^2$ term by expressing h in terms of $\bar{\mu}$. By Lipschitz continuity of ∇f and the coordinate relation $\bar{\mu} = B\mu$ we have

$$\|h\| \le L\|\mu\| = L\|B^\dagger\bar{\mu}\| \le L\|B^\dagger\|\,\|\bar{\mu}\| = \frac{L}{\sigma^+_{\min}(\Xi)}\,\|\bar{\mu}\|, \tag{44}$$

where $B^\dagger = (\Xi \otimes I_n)^\dagger$ and $\sigma^+_{\min}(\Xi)$ denotes the smallest nonzero singular value of Ξ. Therefore

$$\|Bh\| \le \|B\|\|h\| \le \sigma_{\max}(\Xi)\cdot\frac{L}{\sigma^+_{\min}(\Xi)}\,\|\bar{\mu}\| = \sqrt{C}\,\|\bar{\mu}\|, \tag{45}$$

where we define the compact constant

$$C \triangleq \frac{\sigma^2_{\max}(\Xi)}{(\sigma^+_{\min}(\Xi))^2}\,L^2. \tag{46}$$

Substituting (45) into (43) gives

$$\begin{aligned} c_1\dot{V}_{\bar{\mu}} &\le c_1\kappa_1(\epsilon_1+\epsilon_2)\,\|\bar{\mu}\|^2 + \frac{c_1\kappa_1}{4\epsilon_1}\,C\,\|\bar{\mu}\|^2 + \frac{c_1\kappa_1}{4\epsilon_2}\|\bar{\nu}\|^2 \\ &= \left[c_1\kappa_1(\epsilon_1+\epsilon_2) + \frac{c_1\kappa_1 C}{4\epsilon_1}\right]\|\bar{\mu}\|^2 + \frac{c_1\kappa_1}{4\epsilon_2}\|\bar{\nu}\|^2. \end{aligned} \tag{47}$$

This is the desired form. Rewriting norms as quadratic forms yields the final bound

$$c_1\dot{V}_{\bar{\mu}} \le \left[c_1\kappa_1(\epsilon_1+\epsilon_2) + \frac{c_1\kappa_1 C}{4\epsilon_1}\right]\bar{\mu}^\top\bar{\mu} + \frac{c_1\kappa_1}{4\epsilon_2}\bar{\nu}^\top\bar{\nu}, \tag{48}$$

valid for any chosen $\epsilon_1, \epsilon_2 > 0$. The parameters $c_1, \epsilon_1, \epsilon_2$ can then be tuned so that the coefficient multiplying $\bar{\mu}^\top\bar{\mu}$ is made sufficiently small (or negative after combining with other Lyapunov terms) in the overall $\dot{V}$ analysis.

D. Derivative of $V^a_{\bar{\nu}}$ (corrected)

Consider the Lyapunov candidate

$$V^a_{\bar{\nu}} = \tfrac{1}{2}\,\bar{\nu}^\top\bar{\nu} + \sum_{k=1}^{N-1}\frac{w_{k+1,p_k}}{2\kappa_2}\big(\bar{a}_{k+1,p_k}(t) - \phi_{k+1,p_k}\big)^2. \tag{49}$$

Differentiating along system trajectories gives

$$\begin{aligned} \dot{V}^a_{\bar{\nu}} &= \kappa_3\,\bar{\nu}^\top\bar{\mu} - \bar{\nu}^\top(Q^a \otimes I_n)\bar{\nu} - \bar{\nu}^\top(Q \otimes I_n)\bar{\eta} \\ &\quad + \sum_{k=1}^{N-1}\frac{w_{k+1,p_k}}{\kappa_2}\big(\bar{a}_{k+1,p_k} - \phi_{k+1,p_k}\big)\dot{\bar{a}}_{k+1,p_k}. \end{aligned} \tag{50}$$

Expansion of the adaptive term: Using the structure of the DST-based incidence in the reduced coordinates one verifies (cf. previous derivation)

$$\sum_{k=1}^{N-1} w_{k+1,p_k}\bar{a}_{k+1,p_k}\Big(\bar{\nu}_k - \sum_{j+1\in\bar{\mathcal{N}}_{\text{out}}(k+1)}\bar{\nu}_j\Big)^\top\bar{\nu}_k = \bar{\nu}^\top(\bar{Q}^a \otimes I_n)\bar{\nu}, \tag{51}$$

and, with the auxiliary matrix Φ defined as in (**??**), the adaptive contribution combines with the constant part to produce the representation used below. Hence

$$\dot{V}^a_{\bar{\nu}} = \kappa_3\bar{\nu}^\top\bar{\mu} - \bar{\nu}^\top\big((\tilde{Q}^a + \Phi) \otimes I_n\big)\bar{\nu} - \bar{\nu}^\top(Q \otimes I_n)\bar{\eta}, \tag{52}$$

where $\tilde{Q}^a$ denotes the symmetric (or stabilized) part arising from Q^a and the chosen ϕ-shifts.

Young bounds for cross terms: We now bound the cross terms in (52). Fix two positive constants $\epsilon_6 > 0$ and $\epsilon_7 > 0$ (these are free tuning parameters used in Young's inequality). We use the inequality

$$ab \le \frac{a^2}{\epsilon} + \frac{\epsilon}{4}b^2, \qquad \forall a,b \in \mathbb{R},\ \epsilon > 0,$$

which is algebraically equivalent to $(a - \frac{\epsilon}{2}b)^2 \ge 0$.

Apply this to the first cross term $\kappa_3\bar{\nu}^\top\bar{\mu} = \kappa_3\|\bar{\nu}\|\|\bar{\mu}\|$:

$$\kappa_3\|\bar{\nu}\|\|\bar{\mu}\| \le \kappa_3\left(\frac{1}{\epsilon_6}\|\bar{\mu}\|^2 + \frac{\epsilon_6}{4}\|\bar{\nu}\|^2\right). \tag{53}$$

(Note this form yields the coefficients matching the desired final expression.)

For the coupling term with $\bar{\eta}$, use the operator norm $\|Q \otimes I_n\| = \bar{\lambda}(Q)$ and the same type of Young bound:

$$\begin{aligned} -\bar{\nu}^\top(Q \otimes I_n)\bar{\eta} &\le \bar{\lambda}(Q)\|\bar{\nu}\|\|\bar{\eta}\| \\ &\le \frac{\epsilon_7}{2}\|\bar{\nu}\|^2 + \frac{\bar{\lambda}^2(Q)}{2\epsilon_7}\|\bar{\eta}\|^2. \end{aligned} \tag{54}$$

Combine bounds and obtain final form: Substitute (53) and (54) into (52). Collect the terms to obtain

$$\dot V^a_{\bar\nu} \le \kappa_3 \frac{1}{\epsilon_6}\|\bar\mu\|^2 + \left(\kappa_3\frac{\epsilon_6}{4} + \frac{\epsilon_7}{2}\right)\|\bar\nu\|^2 \\ - \bar\nu^\top\big((\tilde Q^a + \Phi)\otimes I_n\big)\bar\nu + \frac{\bar\lambda^2(Q)}{2\epsilon_7}\|\bar\eta\|^2. \tag{55}$$

E. Derivative of $V_{\bar\eta}$

Consider the Lyapunov function

$$V_{\bar\eta} = \frac{1}{2}(\bar\nu+\bar\eta)^\top(\bar\nu+\bar\eta). \tag{56}$$

Taking its derivative along the system dynamics

$$\dot{\bar\nu} = \kappa_3\bar\mu - (Q^a\otimes I_n)\bar\nu - (Q\otimes I_n)\bar\eta, \tag{57}$$
$$\dot{\bar\eta} = (Q^a\otimes I_n)\bar\nu, \tag{58}$$

yields

$$\dot V_{\bar\eta} = (\bar\nu+\bar\eta)^\top(\dot{\bar\nu}+\dot{\bar\eta}). \tag{59}$$

Substituting the dynamics gives

$$\dot V_{\bar\eta} = \kappa_3(\bar\nu+\bar\eta)^\top\bar\mu - \bar\nu^\top(Q\otimes I_n)\bar\eta - \bar\eta^\top(Q\otimes I_n)\bar\eta. \tag{60}$$

The first cross term can be bounded by Young's inequality, for any $\epsilon_8>0$,

$$\kappa_3(\bar\nu+\bar\eta)^\top\bar\mu = \kappa_3\bar\nu^\top\bar\mu + \kappa_3\bar\eta^\top\bar\mu, \tag{61}$$

To separate $\bar\nu$ and $\bar\eta$ contributions, introduce another scalar $\epsilon_{10}>0$:

$$\kappa_3\bar\nu^\top\bar\mu \le \frac{1}{2\epsilon_8}\|\bar\nu\|^2 + \frac{\epsilon_8}{2}(\kappa_3\|\bar\mu\|)^2 \tag{62}$$

$$\kappa_3\bar\eta^\top\bar\mu \le \frac{\kappa_3^2}{\epsilon_{10}}\|\bar\mu\|^2 + \frac{\epsilon_{10}}{4}\|\bar\eta\|^2. \tag{63}$$

For the cross term $-\bar\nu^\top(Q\otimes I_n)\bar\eta$, using the spectral norm and Young's inequality with $\epsilon_9>0$,

$$-\bar\nu^\top(Q\otimes I_n)\bar\eta \le \frac{\bar\lambda(Q^\top Q)}{\epsilon_9}\|\bar\nu\|^2 + \frac{\epsilon_9}{4}\|\bar\eta\|^2. \tag{64}$$

The last term is bounded by the smallest eigenvalue of Q^S:

$$-\bar\eta^\top(Q\otimes I_n)\bar\eta = -\bar\eta^\top(Q^S\otimes I_n)\bar\eta \le -\underline{\lambda}(Q^S)\|\bar\eta\|^2. \tag{65}$$

Combining all bounds, we obtain

$$\dot V_{\bar\eta} \le \left(\frac{\kappa_3^2\epsilon_8}{2} + \frac{\kappa_3^2}{\epsilon_{10}}\right)\|\bar\mu\|^2 + \left(\frac{1}{2\epsilon_8} + \frac{\bar\lambda(Q^\top Q)}{\epsilon_9}\right)\|\bar\nu\|^2 \\ + \left(\frac{\epsilon_{10}}{4} - \underline{\lambda}(Q^S) + \frac{\epsilon_9}{4}\right)\|\bar\eta\|^2. \tag{66}$$

Multiplying both sides by $c_2>0$ gives

$$c_2\dot V_{\bar\eta} \le c_2\left(\frac{\kappa_3^2\epsilon_8}{2} + \frac{\kappa_3^2}{\epsilon_{10}}\right)\|\bar\mu\|^2 \\ + c_2\left(\frac{1}{2\epsilon_8} + \frac{\bar\lambda(Q^\top Q)}{\epsilon_9}\right)\|\bar\nu\|^2 \\ + c_2\left(\frac{\epsilon_{10}}{4} - \underline{\lambda}(Q^S) + \frac{\epsilon_9}{4}\right)\|\bar\eta\|^2. \tag{67}$$

F. Negative Cross Term

We bound the cross term

$$-\alpha\,\dot{(\bar\nu^\top\bar\mu)} = -\alpha\big(\dot{\bar\nu}^\top\bar\mu + \bar\nu^\top\dot{\bar\mu}\big),$$

using the reduced dynamics

$$\dot{\bar\mu} = -\kappa_1(\Xi\otimes I_n)h - \kappa_1\bar\nu, \tag{68}$$
$$\dot{\bar\nu} = \kappa_3\bar\mu - (Q^a\otimes I_n)\bar\nu - (Q\otimes I_n)\bar\eta. \tag{69}$$

Substituting (68)–(69) and expanding gives

$$-\alpha\dot{(\bar\nu^\top\bar\mu)} = -\alpha\kappa_3\|\bar\mu\|^2 + \alpha\,\bar\mu^\top(Q^a\otimes I_n)\bar\nu + \alpha\,\bar\mu^\top(Q\otimes I_n)\bar\eta \\ + \alpha\kappa_1\bar\nu^\top(\Xi\otimes I_n)h + \alpha\kappa_1\|\bar\nu\|^2. \tag{70}$$

We bound each positive (cross) term on the RHS by Young/Cauchy–Schwarz. Let $\epsilon_3,\epsilon_4,\epsilon_5>0$ be free parameters.

1. $\bar\mu^\top(Q^a\otimes I_n)\bar\nu$.
Using $\|Q^a\otimes I_n\| = \lambda^a_{\max}$ and $ab \le \frac{\epsilon_4}{2}a^2 + \frac{1}{2\epsilon_4}b^2$,

$$\alpha\,\bar\mu^\top(Q^a\otimes I_n)\bar\nu \le \alpha\left(\frac{\epsilon_4}{2}\|\bar\mu\|^2 + \frac{(\lambda^a_{\max})^2}{2\epsilon_4}\|\bar\nu\|^2\right). \tag{71}$$

2. $\bar\mu^\top(Q\otimes I_n)\bar\eta$.
Using $\|Q\otimes I_n\| = \lambda^Q_{\max}$ and $ab \le \frac{\epsilon_5}{2}a^2 + \frac{1}{2\epsilon_5}b^2$,

$$\alpha\,\bar\mu^\top(Q\otimes I_n)\bar\eta \le \alpha\left(\frac{\epsilon_5}{2}\|\bar\mu\|^2 + \frac{(\lambda^Q_{\max})^2}{2\epsilon_5}\|\bar\eta\|^2\right). \tag{72}$$

3. $\kappa_1\bar\nu^\top(\Xi\otimes I_n)h$.
Use $\|(\Xi\otimes I_n)h\| \le \sqrt{C}\,\|\bar\mu\|$ from (**??**) and (**??**), then for any $\epsilon_3>0$ the Young form (chosen to match the desired coefficients) gives

$$\alpha\kappa_1\bar\nu^\top(\Xi\otimes I_n)h \le \alpha\kappa_1\|\bar\nu\|\,\|(\Xi\otimes I_n)h\| \\ \le \alpha\kappa_1\sqrt{C}\,\|\bar\nu\|\,\|\bar\mu\| \\ \le \alpha\left(\frac{\kappa_1 C}{2\epsilon_3}\|\bar\mu\|^2 + \kappa_1\frac{\epsilon_3}{2}\|\bar\nu\|^2\right). \tag{73}$$

(We used $xy \le \frac{A}{2\epsilon}x^2 + \frac{\epsilon}{2A}y^2$ with $A=\kappa_1 C$, which is an admissible Young scaling; the displayed form is standard in Lyapunov manipulations.)

4. The $\alpha\kappa_1\|\bar\nu\|^2$ term remains as is.

Now substitute (71)–(73) into (70) and collect like terms:

$$-\alpha\dot{(\bar\nu^\top\bar\mu)} \le \alpha\Big(-\kappa_3 + \frac{\epsilon_4}{2} + \frac{\epsilon_5}{2} + \frac{\kappa_1 C}{2\epsilon_3}\Big)\|\bar\mu\|^2 \\ + \alpha\Big(\frac{(\lambda^a_{\max})^2}{2\epsilon_4} + \kappa_1\frac{\epsilon_3}{2} + \kappa_1\Big)\|\bar\nu\|^2 \\ + \alpha\,\frac{(\lambda^Q_{\max})^2}{2\epsilon_5}\|\bar\eta\|^2. \tag{74}$$

G. *Overall Lyapunov Derivative Bound*

By aggregating the individual bounds of $c_1\dot{V}_{\bar{\mu}}$, $\dot{V}^a_{\bar{\nu}}$, $c_2\dot{V}_{\bar{\eta}}$, and the negative cross term $-\alpha(\bar{\nu}^\top\bar{\mu})$ the upper bounds derived in the previous subsections can be summarized as follows:

$$c_1\dot{V}_{\bar{\mu}} \le \left[c_1\kappa_1(\epsilon_1+\epsilon_2)+\frac{c_1\kappa_1 C}{4\epsilon_1}\right]\|\bar{\mu}\|^2+\frac{c_1\kappa_1}{4\epsilon_2}\|\bar{\nu}\|^2, \tag{75}$$

$$\begin{aligned}\dot{V}^a_{\bar{\nu}} \le &\frac{\kappa_3}{\epsilon_6}\|\bar{\mu}\|^2+\left(\frac{\kappa_3\epsilon_6}{4}+\frac{\epsilon_7}{2}\right)\|\bar{\nu}\|^2\\ &-\bar{\nu}^\top[(\tilde{Q}^a+\Phi)\otimes I_n]\bar{\nu}+\frac{\bar{\lambda}^2(Q)}{2\epsilon_7}\|\bar{\eta}\|^2,\end{aligned} \tag{76}$$

$$\begin{aligned}c_2\dot{V}_{\bar{\eta}} \le\ & c_2\left(\frac{\kappa_3^2\epsilon_8}{2}+\frac{\kappa_3^2}{\epsilon_{10}}\right)\|\bar{\mu}\|^2\\ &+c_2\left(\frac{1}{2\epsilon_8}+\frac{\bar{\lambda}(Q^\top Q)}{\epsilon_9}\right)\|\bar{\nu}\|^2\\ &+c_2\left(\frac{\epsilon_{10}}{4}-\underline{\lambda}(Q^S)+\frac{\epsilon_9}{4}\right)\|\bar{\eta}\|^2,\end{aligned} \tag{77}$$

$$\begin{aligned}-\alpha(\bar{\nu}^\top\bar{\mu})\le\ & \alpha\left(-\kappa_3+\frac{\epsilon_4+\epsilon_5}{2}+\frac{\kappa_1 C}{2\epsilon_3}\right)\|\bar{\mu}\|^2\\ &+\alpha\left(\frac{(\lambda^a_{\max})^2}{2\epsilon_4}+\kappa_1\left(\frac{\epsilon_3}{2}+1\right)\right)\|\bar{\nu}\|^2\\ &+\alpha\frac{(\lambda^Q_{\max})^2}{2\epsilon_5}\|\bar{\eta}\|^2.\end{aligned} \tag{78}$$

By summing (75)–(78), the time derivative of the composite Lyapunov function V satisfies

$$\begin{aligned}\dot{V} &= c_1\dot{V}_{\bar{\mu}}+\dot{V}^a_{\bar{\nu}}+c_2\dot{V}_{\bar{\eta}}-\alpha(\bar{\mu}^\top\bar{\nu})\\ &\le \mathfrak{A}_{\bar{\mu}}\|\bar{\mu}\|^2+\mathfrak{B}_{\bar{\nu}}\|\bar{\nu}\|^2\\ &\quad-\bar{\nu}^\top\left[\left(\tilde{Q}^a+\Phi\right)^S\otimes I_n\right]\bar{\nu}+\mathfrak{C}_{\bar{\eta}}\|\bar{\eta}\|^2,\end{aligned} \tag{79}$$

where the scalar coefficients are explicitly defined as

$$\begin{aligned}\mathfrak{A}_{\bar{\mu}} \triangleq\ & c_1\left[\kappa_1(\epsilon_1+\epsilon_2)+\frac{\kappa_1 C}{4\epsilon_1}\right]+\frac{\kappa_3}{\epsilon_6}+c_2\kappa_3^2\left(\frac{\epsilon_8}{2}+\frac{1}{\epsilon_{10}}\right)\\ &+\alpha\left(-\kappa_3+\frac{\kappa_1 C}{2\epsilon_3}+\frac{\epsilon_4+\epsilon_5}{2}\right),\end{aligned} \tag{80}$$

$$\begin{aligned}\mathfrak{B}_{\bar{\nu}} \triangleq\ & C_1\frac{\kappa_1}{4\epsilon_2}+\frac{\kappa_3\epsilon_6}{4}+\frac{\epsilon_7}{2}+\frac{c_2}{2\epsilon_8}+c_2\frac{\bar{\lambda}(Q^\top Q)}{\epsilon_9}\\ &+\alpha\left(\kappa_1+\frac{\kappa_1\epsilon_3}{2}+\frac{\lambda^2_{\max}(Q^a)}{2\epsilon_4}\right),\end{aligned} \tag{81}$$

$$\mathfrak{C}_{\bar{\eta}} \triangleq \frac{\bar{\lambda}(Q^\top Q)}{2\epsilon_7}+c_2\left(\frac{\epsilon_9+\epsilon_{10}}{4}-\underline{\lambda}(Q^S)\right)+\alpha\frac{\lambda^2_{\max}(Q)}{2\epsilon_5}. \tag{82}$$

Here, $(\tilde{Q}^a+\Phi)^S \triangleq \frac{1}{2}\left[(\tilde{Q}^a+\Phi)+(\tilde{Q}^a+\Phi)^\top\right]$ denotes the symmetric part of $(\tilde{Q}^a+\Phi)$, and $\epsilon_i>0$ $(i=1,\dots,10)$ are arbitrary small scalars introduced to balance cross-term inequalities. The coefficients $\mathfrak{A}_{\bar{\mu}}$, $\mathfrak{B}_{\bar{\nu}}$, and $\mathfrak{C}_{\bar{\eta}}$ characterize the contribution of $\bar{\mu}$-, $\bar{\nu}$-, and $\bar{\eta}$-related components to the Lyapunov derivative. *By appropriate parameter selection, the composite Lyapunov function $\dot{V}$ is rendered negative definite, thereby guaranteeing global asymptotic stability.*

V. Parameters Adjustment

A. *Positive definiteness of the Lyapunov function*

$\alpha^2 < c_1$

B. *Bound of $\mathfrak{A}_{\bar{\mu}}$*

By defining the design parameters

$$\epsilon_{10}=\underline{\lambda}(Q^S),\quad \epsilon_8=\frac{2}{\underline{\lambda}(Q^S)},\quad \epsilon_1=\epsilon_2=1,$$

the upper bound of the term $\mathfrak{A}_{\bar{\mu}}$ can be expressed as

$$\begin{aligned}\mathfrak{A}_{\bar{\mu}} \triangleq\ & C_1\left[\kappa_1(\epsilon_1+\epsilon_2)+\kappa_1\frac{C}{4\epsilon_1}\right]+\frac{\kappa_3}{\epsilon_6}+C_2\kappa_3^2\left(\frac{\epsilon_8}{2}+\frac{1}{\epsilon_{10}}\right)\\ &+\alpha\left(-\kappa_3+\kappa_1\frac{C}{2\epsilon_3}+\frac{\epsilon_4+\epsilon_5}{2}\right).\end{aligned} \tag{83}$$

The always positive components of (83) are

$$C_1\left[\kappa_1(\epsilon_1+\epsilon_2)+\kappa_1\frac{C}{4\epsilon_1}\right]+\kappa_3^2C_2\left(\frac{\epsilon_8}{2}+\frac{1}{\epsilon_{10}}\right).$$

Given that

$$\epsilon_{10}=\underline{\lambda}(Q^S),\quad C_2=\frac{3\bar{\lambda}(Q^\top Q)}{\underline{\lambda}(Q^S)},\quad \epsilon_8=\frac{2}{\underline{\lambda}(Q^S)},$$

it follows that

$$C_2\left(\frac{\epsilon_8}{2}+\frac{1}{\epsilon_{10}}\right)=\frac{6\bar{\lambda}(Q^\top Q)}{\underline{\lambda}^2(Q^S)}.$$

Substituting the above into (83) yields

$$\begin{aligned}\mathfrak{A}_{\bar{\mu}} =\ & C_1\kappa_1\left(\epsilon_1+\epsilon_2+\frac{C}{4\epsilon_1}\right)+\kappa_3\left(\frac{1}{\epsilon_6}-\alpha\right)+\kappa_3^2\frac{6\bar{\lambda}(Q^\top Q)}{\underline{\lambda}^2(Q^S)}\\ &+\alpha\left(\kappa_1\frac{C}{2\epsilon_3}+\frac{\epsilon_4+\epsilon_5}{2}\right)<0.\end{aligned} \tag{84}$$

Letting $\epsilon_1=\frac{\sqrt{C}}{2},\epsilon_3=C$, where $\epsilon_1=\frac{\sqrt{C}}{2}$ guarantees the corresponding term is minimized in this scenario. One obtains

Then, condition (**??**) becomes

$$\begin{aligned}\mathfrak{A}_{\bar{\mu}} =\ & C_1\kappa_1\left(\epsilon_2+\sqrt{C}\right)+\frac{\kappa_3}{\epsilon_6}+\kappa_3^2\frac{6\bar{\lambda}(Q^\top Q)}{\underline{\lambda}^2(Q^S)}\\ &+\alpha\left(\frac{\kappa_1}{2}+\frac{\epsilon_4+\epsilon_5}{2}-\kappa_3\right)<0.\end{aligned} \tag{85}$$

$$\frac{\kappa_1+\epsilon_4+\epsilon_5}{2}<\kappa_3$$

C. *Bound on the Coefficient $\mathfrak{C}_{\bar{\eta}}$*

Recall from (82) that

$$\mathfrak{C}_{\bar{\eta}} \triangleq \frac{\bar{\lambda}(Q^\top Q)}{2\epsilon_7}+c_2\left[\frac{\epsilon_9+\epsilon_{10}}{4}-\underline{\lambda}(Q^S)\right]+\alpha\frac{\lambda^2_{\max}(Q)}{2\epsilon_5}. \tag{86}$$

Letting

$$c_2=\frac{3\,\bar{\lambda}(Q^\top Q)}{\underline{\lambda}(Q^S)},\quad \epsilon_9=\epsilon_{10}=\underline{\lambda}(Q^S), \tag{87}$$

the middle term in (86) becomes

$$c_2\left[\frac{\epsilon_9+\epsilon_{10}}{4}-\underline{\lambda}(Q^S)\right]=\frac{3\,\bar{\lambda}(Q^\top Q)}{\underline{\lambda}(Q^S)}\left[-\underline{\lambda}(Q^S)+\frac{\epsilon_9+\epsilon_{10}}{4}\right] = -\frac{3}{2}\,\bar{\lambda}(Q^\top Q). \tag{88}$$

Therefore, the coefficient $\mathfrak{C}_{\bar{\eta}}$ reduces to

$$\mathfrak{C}_{\bar{\eta}}=\left(\frac{1}{\epsilon_7}-3+\frac{\alpha}{\epsilon_5}\right)\frac{\bar{\lambda}(Q^\top Q)}{2}. \tag{89}$$

In order to determine ϵ_7, with the help of $\bar{\lambda}(Q^\top Q) = (\lambda_{\max}(Q))^2$, we have:

$$\epsilon_7>\frac{1}{3-\frac{\alpha}{\epsilon_5}} \tag{90}$$

$\alpha<3\epsilon_5, lim\alpha<6\epsilon_5$

$$\alpha<3\,\epsilon_5 \tag{91}$$

$$\epsilon_7>\frac{1}{3-\frac{\alpha}{\epsilon_5}} \tag{92}$$

$$\kappa_3>\frac{\kappa_1+\epsilon_4+\epsilon_5}{2} \tag{93}$$

$$C_1\kappa_1(\epsilon_2+\sqrt{C})+\frac{\kappa_3}{\epsilon_6}+\kappa_3^2\frac{6\bar{\lambda}(Q^\top Q)}{\underline{\lambda}^2(Q^S)} +\alpha\left(\frac{\kappa_1}{2}+\frac{\epsilon_4+\epsilon_5}{2}-\kappa_3\right)<0 \tag{94}$$

D. Bound of $\mathfrak{B}_{\bar{\nu}}$

The upper bound of $\mathfrak{B}_{\bar{\nu}}$ can be defined as

$$\mathfrak{B}_{\bar{\nu}}\triangleq C_1\frac{\kappa_1}{4\epsilon_2}+\frac{\kappa_3\epsilon_6}{4}+\frac{\epsilon_7}{2}+C_2\frac{1}{2\epsilon_8}+C_2\frac{\bar{\lambda}(Q^\top Q)}{\epsilon_9} +\alpha\left(\kappa_1+\frac{\kappa_1\epsilon_3}{2}+\frac{\lambda_{\max}^2(Q^a)}{2\epsilon_4}\right), \tag{95}$$

where C_1 and C_2 are the design constants as defined in the previous subsection.

Expanding the constants with typical parameter choices, e.g.,

$$\epsilon_1=\epsilon_2=1,\quad C_2=\frac{3\bar{\lambda}(Q^\top Q)}{\underline{\lambda}(Q^S)},\epsilon_9=\underline{\lambda}(Q^S)$$

the expression can be further simplified to

$$\mathfrak{B}_{\bar{\nu}}=\ \frac{C_1\kappa_1+\kappa_3\epsilon_6}{4}\ +\frac{\epsilon_7}{2}\ +\frac{3}{4}\bar{\lambda}(Q^\top Q)\ +3\left(\frac{\bar{\lambda}(Q^\top Q)}{\underline{\lambda}(Q^S)}\right)^2 +\alpha\left(\kappa_1+\frac{\kappa_1\epsilon_3}{2}\ +\frac{\lambda_{\max}^2(Q^a)}{2\epsilon_4}\right). \tag{96}$$

This bound will be used in the Lyapunov-based stability analysis to ensure the negativity of certain cross terms, and the design parameters ϵ_i, α, C_1, C_2 are chosen to satisfy the sufficient conditions derived from (95) and (96).

1) Adjusting Φ to Dominate $\mathfrak{B}_{\bar{\nu}}$: To ensure that the cross term associated with $\bar{\nu}$ is strictly negative, we consider the matrix

$$-\bar{\nu}^T\left[\left(\tilde{Q}^a+\Phi\right)^S\otimes I_n\right]\bar{\nu},$$

where $\Phi\in\mathbb{R}^{(N-1)\times(N-1)}$ is a tunable parameter matrix designed to dominate $\mathfrak{B}_{\bar{\nu}}$. Each entry of Φ is defined according to the underlying graph structure as

$$\Phi_{kj}=\begin{cases}\phi_{j+1,p_j}w_{j+1,p_j}, & \text{if } j=k,\\ -\phi_{j+1,p_j}w_{j+1,p_j}, & \text{if } j=p_k-1,\\ 0, & \text{otherwise,}\end{cases} \tag{97}$$

where $\phi_{j+1,p_j}>0$ are the adjustable parameters, and w_{j+1,p_j} are the edge weights.

By appropriately choosing the parameters ϕ_{j+1,p_j}, the matrix

$$(\tilde{Q}^a+\Phi)^S-\mathfrak{B}_{\bar{\nu}}\cdot I_{N-1}$$

can be made positive definite. Consequently, the quadratic form

$$-\bar{\nu}^T\left[\left((\tilde{Q}^a+\Phi)^S-\mathfrak{B}_{\bar{\nu}}\cdot I_{N-1}\right)\otimes I_n\right]\bar{\nu} \tag{98}$$

is strictly negative, which guarantees that the $\bar{\nu}$-related term in the Lyapunov derivative is always non-positive, regardless of the magnitude of $\mathfrak{B}_{\bar{\nu}}$.

TABLE I: Key Lyapunov Parameters and Design Constants for the DST System

Parameter	Symbol	Value / Formula
Maximum eigenvalue of $Q^\top Q$	$\lambda_{\max}(Q^\top Q)$	0.343488
Maximum eigenvalue of Q_a	$\lambda_{\max}(Q_a)$	0.406161
Minimum eigenvalue of Q^S	$\lambda_{\min}(Q^S)$	0.026803
Maximum eigenvalue of Q	$\lambda_{\max}(Q)$	0.40891
C constant	C	526696
C_1 constant	C_1	2826
C_2 constant	C_2	38
α	α	$0.5\sqrt{C_1}$
ϵ_3	ϵ_3	C
ϵ_4	ϵ_4	0.1749
ϵ_5	ϵ_5	5.6
ϵ_6	ϵ_6	$2/\alpha$
ϵ_7	ϵ_7	1.0
$\mathfrak{A}_{\bar{\mu}}$	—	Computed from Eq. (83)
$\mathfrak{B}_{\bar{\nu}}$	—	Computed from Eq. (95)
$\mathfrak{C}_{\bar{\eta}}$	—	Computed from Eq. (82)

E. Computation of Lyapunov Bounds and Φ Adjustment

Based on the system matrices and the DST algorithm, the key Lyapunov-related terms are computed as follows:

- $\mathfrak{A}_{\bar{\mu}}$: represents the upper bound of the $\bar{\mu}$-related term in the Lyapunov derivative. It is calculated using

$$\mathfrak{A}_{\bar{\mu}}=A_{\text{term1}}+A_{\text{term2}}+A_{\text{term3}}+A_{\text{term4}}, \tag{99}$$

  where each term is computed from the system parameters C, C_1, C_2, κ_i, ϵ_i, and α as defined in Table I.
- $\mathfrak{B}_{\bar{\nu}}$: corresponds to the cross-term associated with $\bar{\nu}$. It is computed as

$$\mathfrak{B}_{\bar{\nu}}=B_{\text{term1}}+B_{\text{term2}}+B_{\text{term3}}+B_{\text{term4}}+B_{\text{term5}}, \tag{100}$$

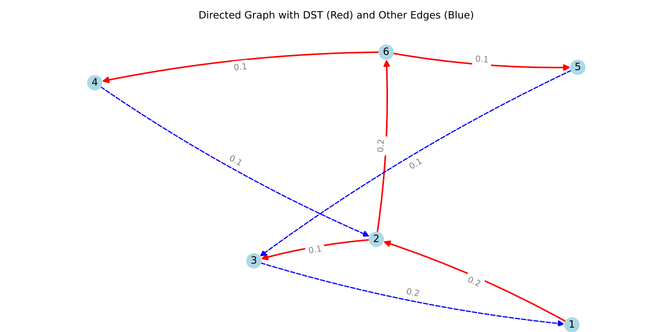


Fig. 1: The communication topology of the 8-agent weight-unbalanced network.

which includes contributions from κ_1, κ_3, ϵ_i, $\lambda_{\max}(Q^a)$, and α.

- $\mathfrak{C}_{\bar{\eta}}$: represents the $\bar{\eta}$ term bound and is calculated from $\lambda_{\max}(Q^\top Q)$, $\lambda_{\max}(Q)$, ϵ_5, ϵ_7, and α.

To guarantee negativity of the $\bar{\nu}$-related term in the Lyapunov derivative, we introduce a tunable matrix $\Phi \in \mathbb{R}^{(N-1)\times(N-1)}$:

$$\Phi_{kj} = \begin{cases} \phi_{j+1,p_j} w_{j+1,p_j}, & j = k, \\ -\phi_{j+1,p_j} w_{j+1,p_j}, & j = p_k - 1, \\ 0, & \text{otherwise}, \end{cases} \tag{101}$$

where ϕ_{j+1,p_j} are adjustable parameters. By iteratively increasing ϕ_{j+1,p_j}, the matrix

$$(\tilde{Q}^a + \Phi)^S - \mathfrak{B}_{\bar{\nu}} \cdot I_{N-1}$$

can be made positive definite. Consequently, the quadratic form

$$-\bar{\nu}^T \left[\left((\tilde{Q}^a + \Phi)^S - \mathfrak{B}_{\bar{\nu}} \cdot I_{N-1} \right) \otimes I_n \right] \bar{\nu}$$

is strictly negative, proving that the $\bar{\nu}$-related term is always non-positive, regardless of the magnitude of $\mathfrak{B}_{\bar{\nu}}$.

VI. Simulation

Consider a distributed system consisting of $N = 8$ agents connected over a directed and weight-unbalanced network. The DST edges are selected as

$$\mathcal{E}_{\text{DST}} = \{(1,2),(1,3),(1,4),(2,5),(2,6),(4,7),(7,8)\},$$

and the fixed feedback edges are

$$\mathcal{E}_{\text{fix}} = \{(6,3),(7,2),(8,1)\}.$$

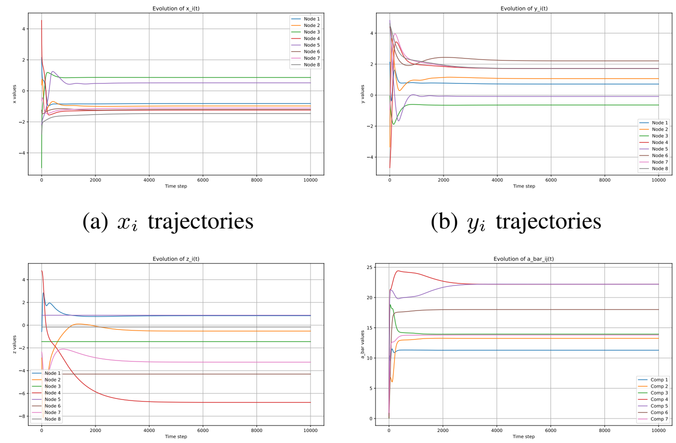
The corresponding edge weights are 0.1 for $\mathcal{E}_{\text{DST}}$ and 0.2 for $\mathcal{E}_{\text{fix}}$.

Each agent i is associated with a nonconvex local cost function

$$f_i(x_i;\xi_i,\zeta_i) = \tfrac{1}{4}\xi_i\|x_i\|^4 + \varepsilon\,\zeta_i^\top x_i,$$

whose gradient is

$$\nabla f_i(x_i;\xi_i,\zeta_i) = \xi_i\|x_i\|^2 x_i + \varepsilon\,\zeta_i.$$



(a) x_i trajectories (b) y_i trajectories (c) z_i trajectories (d) $\bar{a}_{ij}$ adaptive weights

Fig. 2: Evolution of state and adaptive variables under the proposed DST-based dynamics.

The random coefficients ξ_i and ζ_i introduce heterogeneity and perturbations into the cost functions. The Lipschitz constant of ∇f_i is estimated as $L_i = 3\xi_{\max}R^2$ with $R = \sqrt{n}\max(|x_{\min}|,|x_{\max}|)$.

All parameters in the simulation are chosen consistently with the theoretical analysis in Section **??**, and the adaptive gain ϕ_{ij} is iteratively updated until the Lyapunov inequality is satisfied.

A. State and Adaptive Weight Evolution

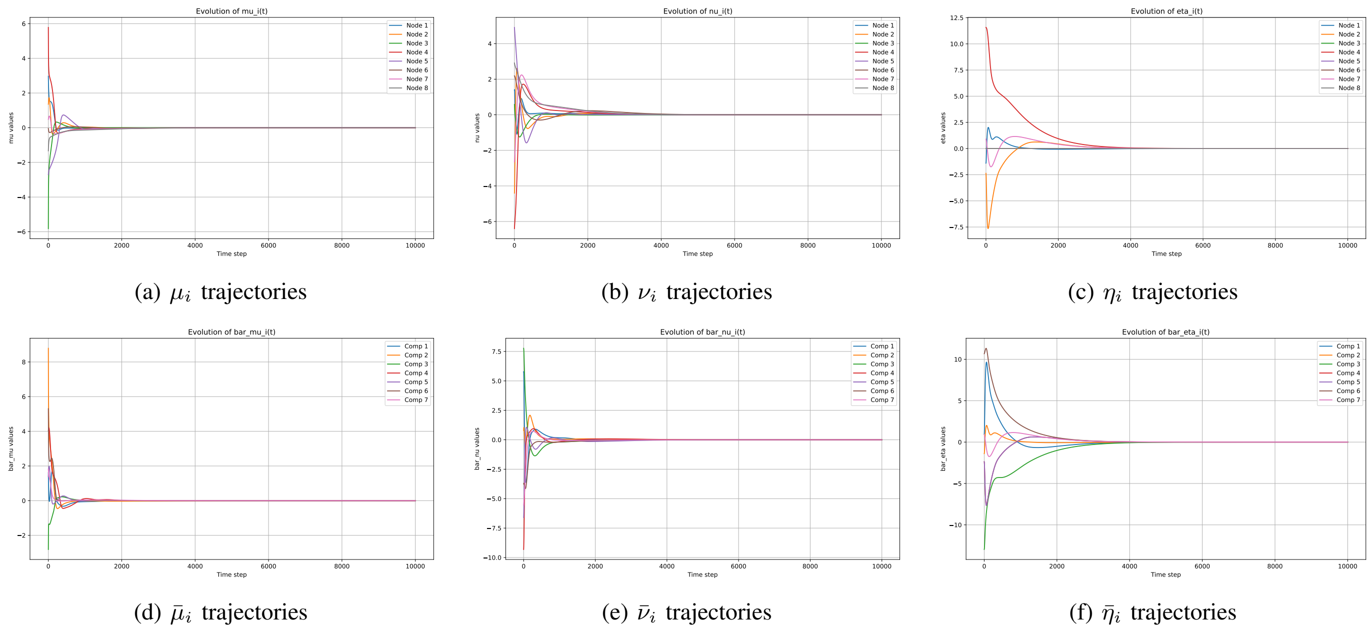
The trajectories of the agent states x_i, auxiliary variables y_i and z_i, and the adaptive edge weights a_{ij} are shown in Fig. **??**. It can be observed that all agents' states x_i achieve consensus under the proposed DST-based adaptive dynamics, despite the directed and weight-unbalanced nature of the communication network. The auxiliary variables y_i and z_i rapidly converge to their steady-state values, indicating the effectiveness of the proposed distributed coordination mechanism. Moreover, the adaptive weights a_{ij} evolve smoothly and remain bounded, confirming that the adaptive law guarantees connectivity maintenance while satisfying the Lyapunov stability condition.

B. Primal–Dual Variable Dynamics

Figure 3 illustrates the temporal evolution of the primal and dual variables (μ_i,ν_i,η_i) and their transformed counterparts $(\bar{\mu}_i,\bar{\nu}_i,\bar{\eta}_i)$. All trajectories converge asymptotically to zero, verifying that the equilibrium point of the closed-loop system coincides with the optimal saddle point of the distributed optimization problem. Notably, the transformed variables exhibit faster decay and smoother transients compared to their original counterparts, which confirms that the proposed coordinate transformation effectively mitigates the influence of network unbalance. These results are consistent with the theoretical stability analysis established in Section **??**.

C. Lyapunov Function Verification

To numerically validate the theoretical results, the evolution of the proposed Lyapunov function and its related quantities



(a) μ_i trajectories (b) ν_i trajectories (c) η_i trajectories

(d) $\bar{\mu}_i$ trajectories (e) $\bar{\nu}_i$ trajectories (f) $\bar{\eta}_i$ trajectories

Fig. 3: Evolution of primal and dual variables under the proposed distributed dynamics. The upper row shows the original variables (μ_i, ν_i, η_i), while the lower row presents their transformed counterparts $(\bar{\mu}_i, \bar{\nu}_i, \bar{\eta}_i)$.

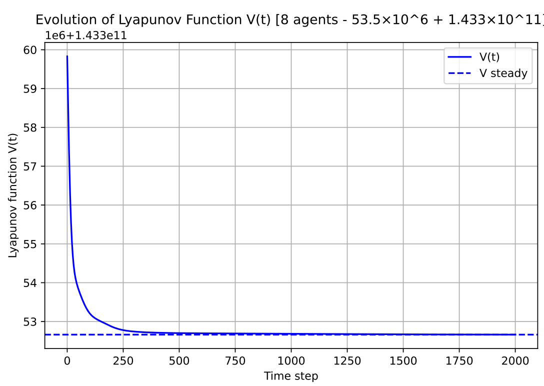


Fig. 4: Evolution of the Lyapunov function $V(t)$, showing monotonic decrease and convergence.

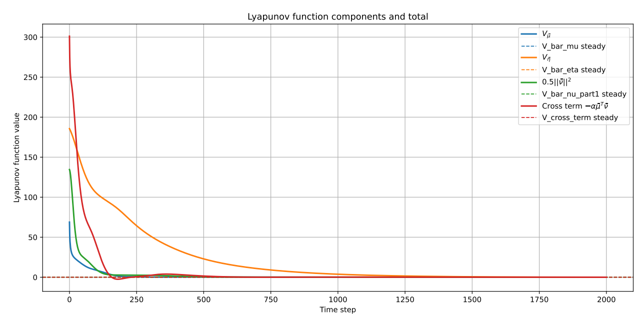


Fig. 5: Component-wise contributions to $V(t)$, where each term remains bounded and converges.

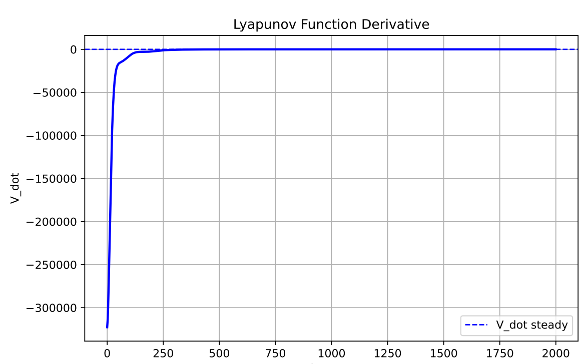


Fig. 6: Derivative $\dot{V}(t)$ along the trajectories, remaining negative after transient oscillations.

are illustrated in Figs. 4–8. Each subfigure independently demonstrates the energy dissipation, component decomposition, and negativity of the derivative and discrete difference, which collectively confirm the asymptotic stability of the closed-loop dynamics.

VII. Conclusion

The conclusion goes here.

Appendix A
Proof of the First Zonklar Equation

Appendix one text goes here.

Appendix B

Appendix two text goes here.

Acknowledgment

The authors would like to thank...

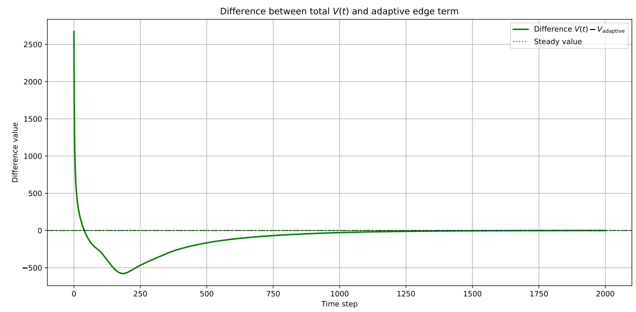


Fig. 7: Discrete difference $\Delta V(t)$, confirming monotonic decrease of $V(t)$ in the discrete-time implementation.

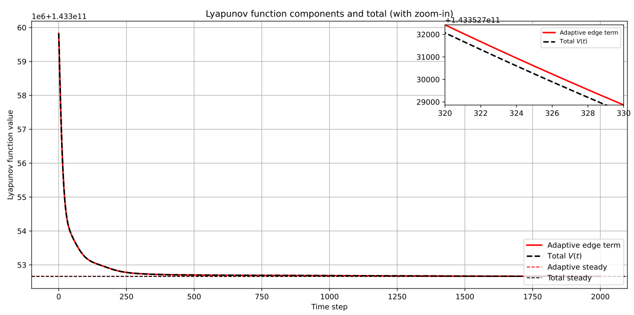


Fig. 8: Discrete difference $\Delta V(t)$, confirming monotonic decrease of $V(t)$ in the discrete-time implementation.

References

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